

Scenerio Based AI Interview Questions

5

Sections

43

Scenarios

∞

Value



Chandra Sekhar

AI Engineer & Trainer

● Full answers + explanations — [Link in caption](#)

5 Topics · 43 Scenario-Based Questions



GenAI Apps using Plain Python

5 scenarios



LangChain

10 scenarios



LangGraph

9 scenarios



RAG (Retrieval-Augmented Generation)

9 scenarios



CrewAI

10 scenarios



SECTION 1 · PART 1 OF 2

GenAI Apps using Plain Python

1 Retrieval Pipeline Failure

You built a RAG chatbot in plain Python (chunking + embeddings + FAISS). Users say answers are wrong even though the information exists in the PDF. How would you debug chunking strategy, embedding quality, retrieval scoring, and final prompt design?

2 Non-Deterministic Model Behavior

Your LLM returns different outputs for the same input in production. What causes this behavior, and how would you make the system deterministic?

3 Structured Output Integrity Breakdown

Your "JSON-only" response still includes extra text. How would you enforce strict JSON formatting and implement safe retry logic?

4 Escalating Embedding Costs

Embedding expenses are rising rapidly. What caching or deduplication strategy would you design, and where would you store embeddings?

- Full answers + explanations — [Link in caption](#)



SECTION 1 · PART 2 OF 2

GenAI Apps using Plain Python

5

Custom Tool-Oriented Agent Design

You need tool-calling without LangChain (calculator, web search, vector search). How would you design the tool schema, argument parsing, and re-invocation loop?

- Full answers + explanations — [Link in caption](#)



LangChain

6

Retrieval Quality Degradation

You built a RAG app using LangChain's RetrievalQA. Users report irrelevant answers. How would you debug retriever configuration, chunking strategy, and prompt template?

7

Incorrect Tool Selection

Your LangChain agent using tools sometimes selects the wrong tool. How would you improve tool descriptions and better control agent reasoning?

8

Memory Saturation Issue

You are using ConversationBufferMemory. After long chats, responses degrade. Why does this happen and which memory strategy would you switch to?

9

Structured Output Instability

Your StructuredOutputParser frequently fails due to formatting errors. How would you make output parsing more reliable?

- Full answers + explanations — [Link in caption](#)



LangChain

10 High Latency in Production

You deployed a LangChain app and latency is high. How would you optimize chains, parallelize calls, or reduce token usage?

11 Hybrid Retrieval Architecture

You need hybrid search (BM25 + vector store) in LangChain. How would you design the retriever pipeline?

12 Agent Infinite Loop Problem

Your agent loops endlessly while using tools. How would you configure max iterations and stopping conditions?

13 Streaming Response Integration

You must add streaming responses in a LangChain app. Where would you enable streaming and how would you handle callbacks?

- Full answers + explanations — [Link in caption](#)



LangChain

14

Observability and Debugging

You need to log intermediate reasoning steps for debugging. How would you use LangSmith or callbacks?

15

Offline Enterprise Deployment

You must deploy LangChain in an enterprise without internet. What components would you replace or self-host?

- Full answers + explanations — [Link in caption](#)



SECTION 3 · PART 1 OF 2

LangGraph

16 Missing Node Execution

You built a multi-step workflow in LangGraph. Sometimes it skips a node. How would you debug state transitions, edges, and conditional checks?

17 Cyclic Execution Loop

Your LangGraph agent loops endlessly between two nodes. How would you design stopping conditions, max hops, and guardrails?

18 Shared State Collision

You are using shared state across nodes. A bug causes overwritten values. How would you structure state updates safely to avoid collisions?

19 Unstable Conditional Routing

You need routing based on LLM output. How would you implement deterministic branching and reduce ambiguity in decisions?

- Full answers + explanations — [Link in caption](#)



SECTION 3 · PART 2 OF 2

LangGraph

20 Node Timeout Resilience

A node fails due to API timeout. How would you implement retries, backoff, and failure recovery inside the graph?

21 Persistent Memory Requirement

You must persist conversation state across sessions. How would you design checkpointing and state rehydration?

22 Parallel Research Orchestration

You want parallel execution of two research nodes before summarization. How would you structure the graph for fan-out and fan-in?

23 Human Approval Gate

You need human-in-the-loop approval before final output. How would you pause execution, capture feedback, and resume the graph?

24 Performance Bottleneck Investigation

Your graph latency is high. How would you profile node-level performance and identify bottlenecks?

• Full answers + explanations — [Link in caption](#)



RAG (Retrieval-Augmented Generation)

25

Partial Answer Accuracy

You built a RAG system, but answers are only partially correct. How would you isolate whether the issue is retrieval quality or generation behavior?

26

Missing Context Retrieval

Users complain the system misses relevant context. How would you redesign chunk size, overlap, and chunk boundaries for better recall?

27

Redundant Retrieval Results

Your retriever returns too many similar chunks. How would you improve diversity in results while keeping relevance high?

28

Similarity Metric Decision

You are choosing between cosine similarity and dot product. How would you decide which to use based on embeddings and indexing setup?

• Full answers + explanations — [Link in caption](#)



RAG (Retrieval-Augmented Generation)

29

Rising Embedding Spend

Embedding costs are rising rapidly. What optimization strategies would you implement to reduce cost without hurting quality?

30

Grounding Failure and Hallucinations

Your RAG system hallucinates beyond retrieved context. How would you restrict generation to use only retrieved data?

31

Multi-Document Attribution

You must support multi-document comparison. How would you maintain reliable source attribution in responses?

32

Context Window Limitations

Long documents exceed context limits. How would you implement hierarchical retrieval or progressive summarization?

• Full answers + explanations — [Link in caption](#)



RAG (Retrieval-Augmented Generation)

33

Explainability Requirement

You need explainability in RAG answers. How would you show which chunks influenced the response and why they were selected?

- Full answers + explanations — [Link in caption](#)



SECTION 5 · PART 1 OF 3

CrewAI

34 Inconsistent Multi-Agent Output

You built a CrewAI system with Researcher and Writer agents. Output quality is inconsistent. How would you redesign roles, goals, and task descriptions to stabilize results?

35 Redundant Agent Contributions

Agents keep producing overlapping content. How would you improve task delegation, boundaries, and reduce duplication?

36 Infinite Reasoning Loops

Your crew enters repeated reasoning loops. How would you enforce stopping rules, max iterations, and convergence checks?

37 Missing Context Handoff

One agent depends on another's output, but context is missing. How would you design shared memory and handoff formats?

- Full answers + explanations — [Link in caption](#)



SECTION 5 · PART 2 OF 3

CrewAI

38

Review Gate Workflow

You must add a Reviewer agent to validate output before final delivery. How would you design the workflow and acceptance criteria?

39

Slow Sequential Execution

Crew execution is slow due to sequential tasks. How would you parallelize independent tasks and merge outputs safely?

40

Tool Integration Safety

You need tool access (web search, calculator). How would you safely integrate tools into CrewAI agents with validation and constraints?

41

Debugging and Observability

You must log intermediate outputs for debugging. How would you implement observability across agents and tasks?

- Full answers + explanations — [Link in caption](#)



SECTION 5 · PART 3 OF 3

CrewAI

42

Hallucination Control

An agent hallucinates facts. How would you introduce validation steps, fact-checking, and confidence gating?

43

Production Deployment Readiness

You must deploy CrewAI in production. How would you manage scaling, retries, failure handling, and error recovery?

- Full answers + explanations — [Link in caption](#)



Want Access to the Entire AI Interview Bundle?

13

COURSES

700+

QUESTIONS

Interview

READY

 INTRODUCING

The AI Interview Master Bundle

13 Interview Courses 650+ Questions



13 Structured Interview Courses

Topic-wise, end-to-end prep



650+ Carefully Curated Questions

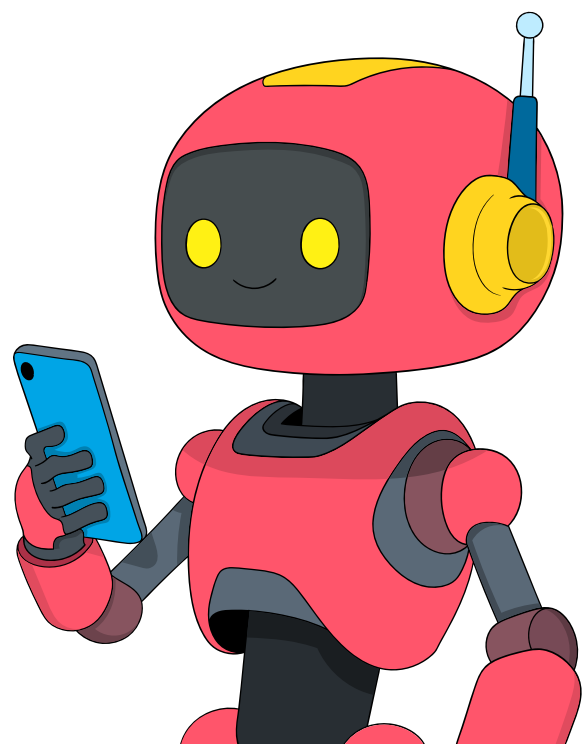
No fluff — interview-ready



Beginner → Advanced

One structured path

Check Next slide for more details



Overview of AI Interview Mastery Bundle

13 modules • 675+ questions

1

Python For GenAI

55 questions

2

Statistics & Math for ML

30 questions

3

Machine Learning Fundam...

70 questions

4

Deep Learning Basics

25 questions

5

LLM Core Concepts

50 questions

6

RAG Interview Mastery

69 questions

7

LangChain Core Concepts

134 questions

8

LangGraph Core Concepts

49 questions

9

Agentic AI

50 questions

10

CrewAI

53 questions

11

AutoGen

28 questions

12

FastAPI (Backend for AI)

32 questions

13

Docker for AI Engineers

30 questions

• Access bundle link from [caption](#)

GET HIRED

The only bundle you'll ever need.



Stop scattered prep. Start structured learning.

13 modules · 645 questions · built for AI engineers



Link of Bundle in caption



Chandra Sekhar

AI Engineer and Trainer

Follow for daily AI interview
questions, tips & career advice.

● Follow on LinkedIn →